

A
Major Project Report on
Event Management System Using Django

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by

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CERTIFICATE

This is to certify that the project entitled “**Event Management System Using Django**” is a bonafide work of Sanika Jawane(22106052), Siddhi Patil(22106122),Mrunal Shejwal (22106132), Shreyas Mahtre(23106135)submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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Project Report Approval

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ABSTRACT

An event management system built with Django typically involves creating a project with a dedicated app for handling events. The system begins with setting up a Django project and app, followed by defining core models such as Event, Attendee, Organizer, and Category to represent various aspects of events. Models are then used to structure data, with forms created to handle user input for event creation and updates. Views are developed to manage the display, creation, and editing of events, while URLs route user requests to the appropriate views. Templates are crafted to present the data in a user-friendly format. To manage events through the Django admin interface, models are registered. Finally, migrations are applied to set up the database schema, and the development server is launched to test the system. Additional features such as user authentication, search functionality, and event registration can be integrated to enhance the system's capabilities.

To make the system functional, views and URLs are implemented to handle requests and direct users to the appropriate pages, such as event lists, details, and forms for adding or editing events. Templates are used to render these pages in a user-friendly manner. The Django admin interface is configured to allow easy management of events and related data. After setting up the models, views, and templates, migrations are applied to establish the database schema, and the development server is launched for testing. Additional features like user authentication and event registration can further enhance the system, providing a comprehensive solution for managing events.

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CHAPTER 1

INTRODUCTION

1.INTRODUCTION

An Event Management System (EMS) is designed to streamline the complex processes involved in organizing and managing events, from scheduling and registrations to ticketing and communication. By utilizing an EMS, organizers can enhance efficiency, reduce administrative burden, and ensure a smooth experience for both attendees and planners.

Django, a high-level Python web framework, is an excellent choice for developing such a system. Known for its simplicity, security features, and rapid development capabilities, Django provides the essential tools needed to build a robust and scalable EMS. Its built-in components, such as user authentication and an administrative interface, facilitate quick deployment and management of event-related tasks, making it an ideal platform for creating an effective and user-friendly Event Management System.

An Event Management System (EMS) is a software solution designed to facilitate the planning, organization, and execution of events. It addresses various logistical challenges such as managing guest lists, handling registrations, scheduling, and coordinating resources. By integrating these functions into a single platform, an EMS helps streamline workflows, minimize errors, and improve the overall efficiency of event management. Traditional methods, often reliant on manual processes and disparate tools, can be time-consuming and prone to mistakes. A well-designed EMS automates these tasks, providing a centralized system that enhances productivity and user experience.

Django, a popular high-level Python web framework, is well-suited for developing an EMS due to its comprehensive feature set and emphasis on simplicity and security. Django follows the "batteries-included" philosophy, offering built-in components that accelerate the development process. Key features include an admin interface for easy management, an authentication system for secure user access, and tools for rapid database integration. This allows developers to focus on customizing the system to meet specific event management needs rather than reinventing basic functionalities.

One of Django's notable advantages is its robust security framework. Django includes protections against common web vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). For an EMS, which handles sensitive data such as personal information and payment details, these security features are crucial. Django's commitment to secure coding practices ensures that the system remains safe from potential threats, giving users confidence in the protection of their data. The scalability of Django is another significant benefit. Whether managing a small community event or a large international conference, Django's architecture supports growth and can handle increasing demands.

. Its modular design allows for the easy addition of new features and functionalities as requirements evolve. This flexibility ensures that the EMS can adapt to different scales and complexities, making it a versatile solution for various types of events.

Finally, Django's strong community support and extensive documentation contribute to its effectiveness as a development framework. The active community provides a wealth of resources, including tutorials, plugins, and forums, which are invaluable for troubleshooting and enhancing the EMS. Comprehensive documentation further aids in understanding and utilizing Django's features, ensuring that developers can implement and maintain a high-quality Event Management System efficiently. This support ecosystem not only speeds up development but also helps in keeping the system updated and aligned with best practices.

Django also benefits from extensive community support and comprehensive documentation. The large and active Django community provides valuable resources such as forums, tutorials, and third-party libraries, which can aid in the development and maintenance of the EMS. The framework's documentation is detailed and well-organized, offering guidance on everything from basic setup to advanced features. This support network helps developers overcome challenges and stay updated with best practices, contributing to the successful implementation and continuous improvement of the Event Management System.

CHAPTER 2

LITERATURE SURVEY

1. LITERATURE SURVEY

2.1-HISTORY

COVID-19 (Coronavirus) has affected day to day life and is slowing down the global economy. This pandemic has affected thousands of peoples, who are either sick or lost their soul due to the spread of this disease The pandemic is fast-moving, as are the efforts to address it. Hospitals reported that their most significant challenges centered on testing and caring for patients with known or suspected COVID-19 and keeping staff safe. Hospitals also reported substantial challenges maintaining or expanding their facilities' capacity to treat patients with COVID-19. Capacity concerns emerged as hospitals anticipated being overwhelmed if they experienced a surge of patients, who may require special beds and rooms to treat and contain infection. Many hospitals reported that post-acute-care facilities were requiring negative COVID-19 tests before accepting patients discharged from hospitals, meaning that some patients who no longer required acute care were taking up valuable bed space while waiting to be discharged.

Hospitals reported that shortages of critical supplies, materials, and logistic support that accompany more beds affected hospitals' ability to care for patients. Hospitals reported needing items that support a patient room, such as intravenous therapy (IV) poles, medical gas, linens, toilet paper, and food. Others reported shortages of no-touch infrared thermometers, disinfectants, and cleaning supplies. Isolated and smaller hospitals faced special challenges maintaining the supplies they needed and restocking quickly when they ran out of supplies.

The other most prominent concerns reported by hospital administrators centered on maintaining facility operations while receiving and treating patients with known or suspected cases of COVID-19. These challenges included concerns about bed availability, particularly specialized beds such as intensive care unit beds, and supplies, as well as maintaining financial solvency given reductions in routine patient care and elective surgeries.

Hospitals anticipated being overwhelmed by a surge in COVID-19 patients, who would need specialty beds and isolation areas for effective treatment. Specifically, hospitals reported concerns about potential shortages of intensive care unit beds, negative pressure rooms, and isolation units. Hospitals also reported that, given the limitations to bed availability, it was challenging to sufficiently separate COVID-19 and non-COVID-19 patients within their facilities.

Hospitals reported they do not have a reliable source for the equipment and supplies they use to support patient care. One hospital reported that, in addition to beds, it needed to source the materials that accompany additional beds and did not know where to order them. For example, hospitals described the supplies that support a patient room, such as intravenous therapy poles, medical gas, linens, and food.

2.2-LITERATURE REVIEW

Life Care GPS based Medical Emergency Solution (IEEE EXPLORE 2020) Govinda Gindodia, Deepali Shrikhande

This application, Life Care, will help a patient to find a specialized doctor as per their needs, availability, distance, and consultancy charges. It is designed especially for an emergency and reduce the patient's time in hunting for a specialized doctor. Life Care will locate the nearby pharmacy store and check the medicines available in the store. This system will digitize the entire medical documentation that can be accessed by doctors and patients from anywhere at any time. Our system provides a link to virtual communication between patients and doctors. The feature of GPS based medical emergency services will help the user to find the closest doctor through GPS.

Life Care Careggi Smart Hospital (IEEE EXPLORE 2020) A. Luschi,A .Belardinelli,L. Marzi,F. Frosini,R. Miniati, E. Iadanza

The application is designed for Android smartphones and tablets and it is freely downloadable from the Google Play Store. It provides various useful tools to hospital's users such as personnel and structures finding, way finding and the possibility to access personal medical records collected on regional electronic health record.

Elderly Healthcare Assistance Application using Mobile Phone (IEEE EXPLORE 2017) Andreas Handojo, Tioe Julio, Adrian Sutiono, Anita Nathani Purbowo

As people become older they need special health, especially from the family member or doctors. Therefore, this research try to develop an application on mobile phone that could help elderly people and their family member to supervise and monitor the health of the elderly. This application has feature to monitor the location of the elderly, remainder to take the medication, doctor appointment remainder, medical record records, emergency phone to family number or personal doctor.

COWAR An Android Based Mobile Application to Help Citizens and COVID-19 Warriors (IEEE EXPLORE 2020) Tarun Saxena, Prince Anuragi, Gandhali Shinde, Nitesh Yadav, Mayuri Digalwa

In this paper, the design and development an android application to spread awareness and to help the people of the world amid this COVID-19. The COWAR app connects the people with the Doctors and the administration to come together and fight the pandemic. With this app, one can track the spread of the COVID-19 epidemic, check live statistics, check for symptoms, browse an interactive map, and also get prevention details.

Blood Donation And Life Saver App (IEEE EXPLORE 2017) Annish Brislin M R1, Albert Mayan J2 , Aroul Canessane R3, Anish Hamlin M R4

It can check for blood donor nearby by using GPS. Once the app user enters the blood group which he/she needed it will automatically show the donor nearby and send an alert message to the donor. In case if the first donor is not available it will automatically search the next donor which is present in queue. If the donor accepts the request, then a onetime password (OTP) will be sent to the donor to verify. Blood donation app provider list of donors in your city/area.

Inside Me: A Proposal for Healthcare MobileApplication (IEEE EXPLORE 2016) Sarawut Busssadee, Sittipong Suwannatria Arnon Chonrawut, Ek Thamwiwatthan Kitsuchart Pasupa

In this paper, we proposed mobile application for health monitoring-Inside Me-which can help users to become more aware of their health. This application aims to track user's workout activities and monitor and analyze user's health condition. It also gives some instructions and suggestions to the user for maintaining and improving his or her health. Moreover, it provides an assessment of the risk that the user may have one of these two diseases: coronary disease and diabetes. Input data are collected from several sources such as questionnaire, medical check-up record, and wearable device.

Health Channel A Health Care Support App(IRJET JOURNAL 2018) Mr. Sujith JohnMs. Dhikhi TMr. Karthikeyan S. Mr. Advait N. Menon

The developed a web app that stores patient and hospital records online on a cloud server accessible to both the doctor and patient. It provides various useful tools to the hospital's users such as personnel directory, access to personal medical records collected on regional electronic health record. The web application keeps track of the doctor's prescription and reorders the

medicines for the patient so that the patient does not go out of stock ever and the doctor can make changes to his prescription.

Mobile App to Search for Nearest Hospital and Healthcare Management System (IJATIR 2016) Pallavi R, Kumar Rajat

The proposed framework finds closest accessible doctor's facility, contacts its rescue vehicle crisis framework, gets to an Electronic Wellbeing Record of crisis patient that can fundamentally help with pre-clinic medicines. The framework will recognize accessibility of the closest accessible particular clinic every through CMS server which gives constant data about the approaching patient to the doctor's facility. This paper proposes Android Based Following for CMS (Crisis Medicinal Framework) on cloud

Developing a mobile COVID-19 prototype management application

(IEEE EXPLORE 2020) Mouna Berquedich, Amine Berquedich, Oulaid Kamach, Malek Masmoudi, Ahmed Chebbak, Laurent Deshayes

COVID-19 is a novel coronavirus that has affected an unprecedented number of people to date. Patients typically present with a combination of fever or cough and have a history of exposure to either a close contact with COVID-19 or travel to an affected geographic area. The applications implemented are used to early recognition and isolation of a patient with COVID-19 at home and in the ED may help decrease exposure to other patients and healthcare personnel. Future research is necessary to expand our collective knowledge of COVID-19 and optimize patient outcomes.

Health technology management: A database analysis as support of technology managers in hospitals Roberto Miniati, Fabrizio Dorib, Ernesto Iadanzac, Mario M. Fregonarad Guido Biffi Gentile

Technology management in healthcare must continually respond and adapt itself to new improvements in medical equipment. Multidisciplinary approaches which consider the interaction of different technologies, their use and user skills, are necessary in order to improve safety and quality. An easy and sustainable methodology is vital to Clinical Engineering (CE) services in healthcare organizations in order to define criteria regarding technology acquisition and replacement. This article underlines the critical aspects of technology management in hospitals by providing appropriate indicators for benchmarking CE services exclusively referring to the maintenance database from the CE department at the Careggi Hospital in Florence, Italy.

CHAPTER 3

Problem Statement

3.Problem Statement

The organization and execution of events involve numerous complex tasks such as scheduling, registration, ticketing, and communication, which are often managed through inefficient, manual processes and disparate tools. These traditional methods can lead to errors, miscommunications, and administrative burdens, particularly when handling large-scale or multiple events. The challenges include coordinating event details, managing attendee registrations and payments, ensuring effective communication, and handling data securely. These issues can hinder the overall efficiency and success of events, causing potential delays and frustrations for both organizers and participants.

To address these problems, the development of an Event Management System (EMS) using Django is proposed. Django, a high-level Python web framework, provides a comprehensive and integrated solution that streamlines event management processes. By centralizing functions such as event scheduling, registration, ticketing, and communication into a single platform, Django's EMS can reduce manual errors, improve data management, enhance security, and facilitate real-time reporting and analytics. This approach ensures a more efficient, scalable, and user-friendly system for organizing and executing events, ultimately leading to better outcomes and experiences for all stakeholders involved.

In this description students have to write Configuration of computer system to run the software.

CHAPTER 5

Proposed System & Implementation

3. Proposed system & Implementation

5.1 Block diagram of proposed system

Students have to draw block diagram of proposed system. Flow chart can be added at this point.

5.2 Description of block diagram

Explanation of block diagram.

5.3 Implementation

Implementation of proposed system must be included here. Students can explain implementation using screen shots of output.

5.4 Advantages/ Application/ result table can be included in this subsection.

CHAPTER 6

Conclusion

4. Conclusion

Students have to include conclusion here. Future scope can be include in next subpoint.

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Research paper

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